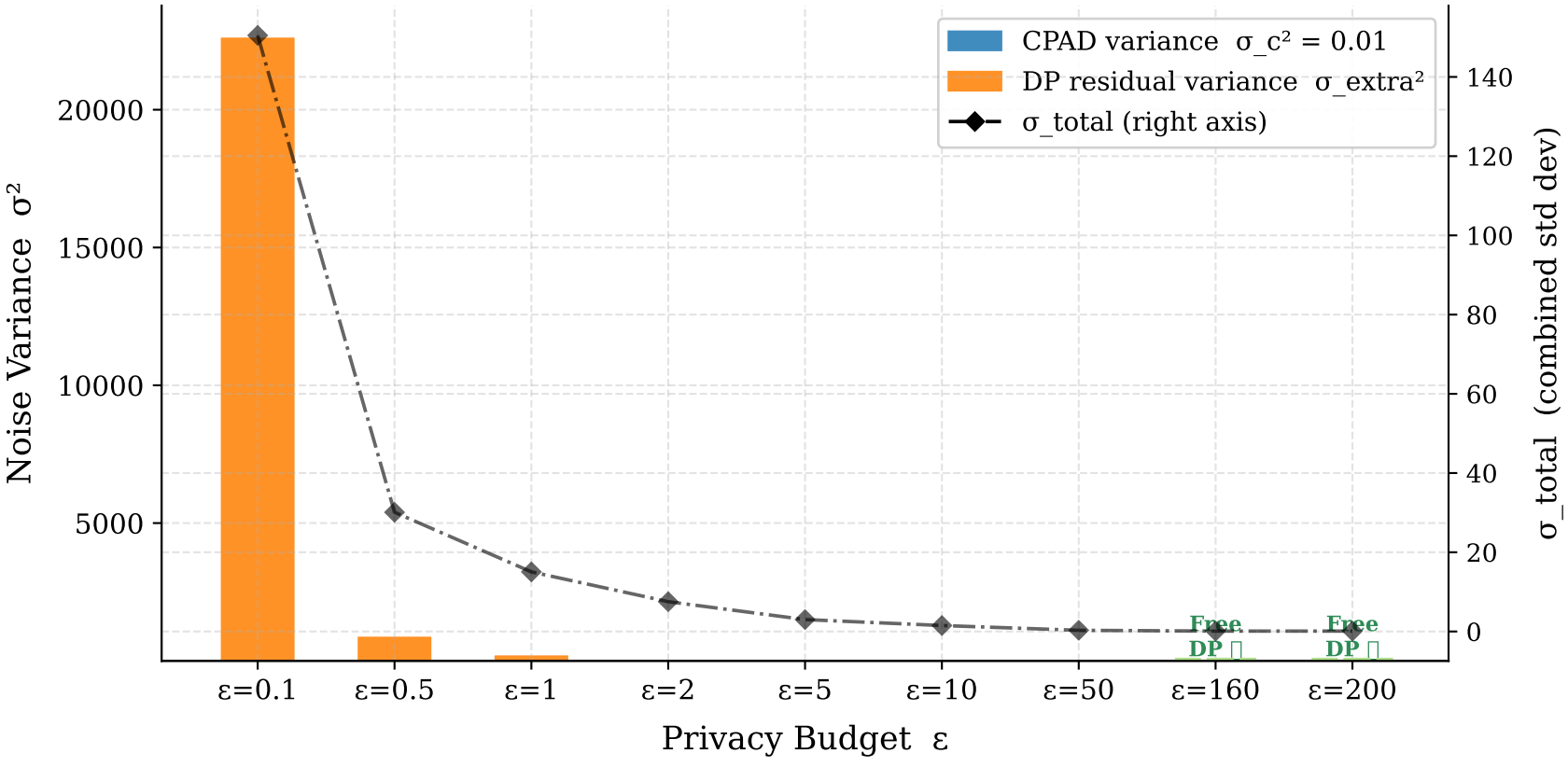
Figure 2 · Noise Budget Decomposition:
Cryptographic (CPAD) vs. Algorithmic (DP) Variance
CPAD variance σ_c² = 0.01
DP residual variance σ_extra²
σ_total (right axis)
Noise Variance σ²
σ_total (combined std dev)
20000
15000
10000
5000
140
120
100
80
60
40
20
0
ε=0.1
ε=0.5
ε=1
ε=2
ε=5
ε=10
ε=50
ε=160
ε=200
Free DP
Free DP
Privacy Budget ε